

Muhammad Zrar Akbar

Computer Science Undergraduate | Python Developer | AI & Machine Learning Enthusiast

sahiwal, Punjab, Pakistan
zrarakbar1@gmail.com | +923226614394

GitHub: [M.Zrar Akbar](#)

LinkedIn: [M.Zrar Akbar](#)

PROFESSIONAL SUMMARY

Computer Science Undergraduate specializing in Python, Data Preprocessing, and Machine Learning Workflows. Demonstrated experience in developing rule-based AI engines, feature extraction pipelines, and predictive prioritization models using Pandas and NumPy. Strong technical foundation in Object-Oriented Programming and Data Structures, focused on building scalable data-driven systems.

EDUCATION

Bachelor of Science in Computer Science

COMSATS University Islamabad, Sahiwal Campus Bachelor of Science in Computer Science (BSCS) | Expected Graduation: 2027 Semester: 5th | CGPA: 2.82 / 4.00

TECHNICAL SKILLS

- Programming Languages:** Python, C++, Java, SQL
- Machine Learning & Data Science:** NumPy, Pandas, Scikit-Learn, Datasets, Data Preprocessing, Feature Engineering
- Tools & Platforms:** Git, GitHub, MySQL, Firebase, Flutter, VS code, IntelliJ IDEA, PyCharm,
- Core Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems

CERTIFICATIONS & PROJECTS

FlyRank Content Refresh Prioritization Model | Python, Pandas, NumPy, Scikit-Learn, Datasets

- Problem & Scope:** Targeted web pages experiencing organic search decay to automate content update prioritization using historical ranking datasets.
- Technical Implementation:** Cleaned, transformed, and engineered time-decay features across search position metrics using Pandas and NumPy data pipelines. Trained Scikit-Learn classification models to identify target pages.
- Outcome & Impact:** Achieved 84% predictive accuracy on validation test splits, establishing clear metrics for high-priority content updates.

MediBot (Medical Knowledge & Rule-Based AI Engine) | Python, JSON, Datasets, Rule-Based AI

- Problem & Scope:** Developed a structured medical guidance chatbot to automate initial symptom query parsing and rule matching.
- Technical Implementation:** Ingested and structured JSON medical datasets into modular dialogue states, utilizing Python dictionary indexing for intent classification.
- Outcome & Impact:** Reduced intent-matching processing latency by 35% compared to baseline nested logical checks across 50+ test evaluation runs.

Arduino Motion Optimization & Preprocessing Engine | Python, G-Code Parsing, Vector Processing

- Problem & Scope:** Solved trajectory jitter and execution delays in an Arduino Nano 2D CNC plotter during complex vector rendering.
- Technical Implementation:** Built Python data preprocessing scripts to parse raw geometric coordinates and strip redundant trajectory control points.
- Outcome & Impact:** Streamlined G-code file sizes by 28%, significantly improving real-time motor response times and lowering path execution drift.

SkillLink (Labor Service Marketplace Web Platform) | Web Architecture, MySQL, Firebase

- Problem & Scope:** Built a digital platform to connect daily wage laborers and loaders with local service opportunities.
- Technical Implementation:** Designed normalized relational MySQL database schemas, secure session workflows, and role-based access controls for service management.
- Outcome & Impact:** Successfully deployed web service endpoints ensuring sub-second query response times for profile

SOFT SKILLS & LANGUAGES

- Soft Skills:** Problem Solving, Analytical Thinking, Collaboration, Adaptability, Time Management
- Languages:** English, Urdu